

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET G — FINAL MASTERY & GRAND CHALLENGE TEST (60 MINS)****CHAPTER 03: ATMOSPHERE AND CLIMATE**

TIME ALLOWED: 60 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-G-03

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET G - Final Mastery & Grand Challenge Test):

1. The question paper comprises **25 Olympiad / NTSE / Advanced Analytical Multiple Choice Questions** of **1 mark each**.
2. Designed for **Final Curricular Mastery & In-Depth Analytical Rigour**: Target time is **60 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **GRAND CHALLENGE SPECIAL:** This final assessment tests your multi-disciplinary analytical reasoning across historical source criticism, geomorphological dynamics, constitutional jurisprudence, and microeconomic market equilibria!

Q1. Using Stefan-Boltzmann's law of radiation ($E = \sigma T^4$), what would Earth's mean surface equilibrium temperature be in the COMPLETE ABSENCE of atmospheric greenhouse gases?

- ☐ A +15°C ☐ B 0°C ☐ C -18°C (approx 255 Kelvin) ☐ D -100°C

Scholarly Concept & Analysis: Without the natural greenhouse effect, radiative equilibrium with solar insolation yields an effective emission temperature of 255 K (-18°C).

Q2. When an unsaturated air parcel ascends adiabatically, its temperature drops at the Dry Adiabatic Lapse Rate (DALR) of:

- ☐ A 1°C per 1,000 m ☐ B 6.5°C per 1,000 m ☐ C 9.8°C per 1,000 m (~10°C/km) ☐ D 20°C per 1,000 m

Scholarly Concept & Analysis: The dry adiabatic lapse rate ($\frac{g}{c_p}$) is approximately 9.8°C per 1,000 meters of elevation change.

Q3. Under what condition is an atmospheric layer classified as 'Absolutely Unstable' for convective cloud development?

- ☐ A When Environmental Lapse Rate is less than Saturated Adiabatic Lapse Rate (ELR < SALR) ☐ B When Environmental Lapse Rate is greater than Dry Adiabatic Lapse Rate (ELR > DALR) ☐ C When temperature is identical at all altitudes ☐ D When the air parcel is 100% frozen solid

Scholarly Concept & Analysis: When the environmental lapse rate exceeds the DALR, an ascending air parcel remains warmer and less dense than surrounding air, accelerating convection.

Q4. What is the 'Geostrophic Wind' in dynamic meteorology?

- ☐ A A hot desert wind blowing along river channels ☐ B A theoretical wind resulting from an exact balance between the horizontal Pressure Gradient Force and the Coriolis Force ☐ C A wind blowing directly into the ground ☐ D A sea breeze blowing during twilight

Scholarly Concept & Analysis: Geostrophic balance occurs above the friction layer when Coriolis force balances the pressure gradient force, flowing parallel to isobars.

Q5. In the Southern Oscillation Index (SOI), what atmospheric pressure difference is measured to assess the El Niño / La Niña state?

- ☐ A Pressure difference between London and New York ☐ B Surface air pressure anomaly difference between Tahiti (Central Pacific) and Darwin, Australia (Western Pacific) ☐ C Pressure difference between the North Pole and South Pole ☐ D Pressure difference between Mumbai and Delhi

Scholarly Concept & Analysis: The Southern Oscillation Index (SOI) measures normalized pressure anomalies between Tahiti and Darwin; sustained negative SOI indicates El Niño.

Q6. What is the Indian Ocean Dipole (IOD), and how does a 'Positive IOD' phase influence the Indian monsoon?

- ☐ A It brings freezing blizzards to Gujarat ☐ B A positive IOD features warmer sea surface temperatures in the western Indian Ocean (near Arabian Sea), enhancing moisture and monsoon rainfall over India ☐ C It turns the Indian Ocean into freshwater ☐ D It causes all monsoon clouds to reverse to Africa

Scholarly Concept & Analysis: A positive IOD warms the western Indian Ocean relative to the east, reinforcing convective cells and boosting Indian monsoon rainfall.

Q7. What is the Madden-Julian Oscillation (MJO)?

- ☐ A A stationary volcanic hot spot in the Pacific ☐ B An eastward-moving global convective pulse of cloudiness and rainfall in the tropical atmosphere that recurs every 30 to 60 days ☐ C An annual polar hurricane in Antarctica ☐ D A local sea breeze on the coast of France

Scholarly Concept & Analysis: The MJO is an eastward-propagating tropical wave of enhanced rainfall that modulates monsoon active-break cycles on 30–60 day timeframes.

Q8. Why does a mature tropical cyclone require low vertical wind shear (< 10 knots) between the lower and upper troposphere to survive?

- ☐ A High wind shear tears apart the vertically stacked thermal core of cumulonimbus towers, dissipating the cyclone's latent heat engine ☐ B Wind shear heats the ocean to boiling temperatures ☐ C Wind shear creates too much rain in the eye ☐ D Wind shear forces the storm to cross the equator

Scholarly Concept & Analysis: Strong vertical wind shear tilts and disperses the warm latent heat core, disrupting the chimney effect and tearing the vortex apart.

Q9. What is the 100-year Global Warming Potential (GWP) of Methane (\$CH_4\$) relative to Carbon Dioxide (\$CO_2\$)?

- ☐ A 1 (identical to CO₂) ☐ B Approximately 28 to 36 times more potent than CO₂ ☐ C 0.01 ☐ D 1,000,000

Scholarly Concept & Analysis: Over a 100-year time horizon, methane traps approximately 28 to 36 times more infrared radiation per kilogram than \$CO_2\$.

Q10. What gas has the highest known Global Warming Potential (GWP), exceeding 23,500 over a 100-year timeframe, used as an electrical insulator in transformers?

- ☐ A Nitrous oxide (\$N_2O\$) ☐ B Sulfur hexafluoride (\$SF_6\$) ☐ C Methane (\$CH_4\$) ☐ D Carbon monoxide (\$CO\$)

Scholarly Concept & Analysis: Sulfur hexafluoride (\$SF_6\$) is an extraordinarily potent greenhouse gas with a GWP of \$\approx 23,500\$ and an atmospheric lifetime of 3,200 years.

Q11. What is the 'Ice-Albedo Positive Feedback' in global climate dynamics?

- ☐ A Melting reflective white ice lowers surface albedo, exposing dark ocean water that absorbs more solar heat, accelerating further ice melt ☐ B Ice reflects 100% of heat into outer space forever ☐ C Melting ice causes global volcanic eruptions ☐ D Ice increases the temperature of the stratosphere

Scholarly Concept & Analysis: Ice loss replaces high-albedo ice with low-albedo ocean water, driving greater absorption of insolation and compounding warming.

Q12. What thermodynamic parameter determines the maximum moisture-holding capacity of an air parcel as expressed by the Clausius-Clapeyron relation?

- ☐ A Atmospheric temperature—water-holding capacity increases by approximately 7% per 1°C rise in temperature ☐ B Atmospheric magnetic flux ☐ C Soil nitrogen percentage ☐ D Speed of Earth's orbital revolution

Scholarly Concept & Analysis: The Clausius-Clapeyron equation dictates that the atmosphere's saturation vapor pressure expands exponentially by ~7% per °C of warming.

Q13. What is the primary thermodynamic cause of the 'October Heat' phenomenon experienced in peninsular India following the Southwest Monsoon?

- ☐ A Sudden emergence of desert sandstorms ☐ B Clear skies and high insolation combined with saturated, high-humidity wet soils, creating oppressive sultry conditions ☐ C Subterranean volcanic eruptions under Mumbai ☐ D A total eclipse of the Moon

Scholarly Concept & Analysis: The transition period in October features clear sunny skies over soil still saturated with monsoon moisture, producing sultry 'October heat'.

Q14. The 'Rossby Waves' (planetary waves) in the upper troposphere develop primarily due to:

- ☐ A Tidal gravitational pull of the moon ☐ B The latitudinal variation of the Coriolis parameter ($\partial f / \partial y$) and planetary thermal gradient between poles and equator ☐ C Oceanic waves splashing into clouds ☐ D Industrial smoke emissions from factories

Scholarly Concept & Analysis: Rossby waves arise from conservation of potential vorticity and the meridional gradient of the Coriolis force (β -effect).

Q15. Why does the Subtropical Westerly Jet stream withdraw to the north of the Himalayas before the onset of the Indian summer monsoon?

- ☐ A The jet stream freezes solid and stops moving ☐ B Summer heating over the Tibetan Plateau and North India displaces the thermal gradient northward, allowing the Tropical Easterly Jet to establish over the peninsula ☐ C The jet stream is blown into outer space by typhoons ☐ D The Himalayas grow taller during summer

Scholarly Concept & Analysis: The northward shift of the subtropical westerly jet beyond Tibet is a mandatory dynamic precursor to the onset of the Indian southwest monsoon.

Q16. What is 'Rayleigh Scattering' in atmospheric optics, and what everyday optical phenomenon does it explain?

- ☐ A Scattering of electromagnetic radiation by particles much smaller than the wavelength, explaining why the clear daytime sky appears blue ☐ B Reflection of radar waves by airplanes ☐ C Absorption of ultraviolet rays by stratospheric ozone ☐ D The formation of desert mirages by hot sand

Scholarly Concept & Analysis: Rayleigh scattering intensity is proportional to λ^{-4} ; short blue wavelengths scatter much more than longer red wavelengths, coloring the sky blue.

Q17. What optical scattering mechanism causes clouds and fog droplets (which are larger than light wavelengths) to appear white?

- ☐ A Rayleigh scattering ☐ B Mie scattering ☐ C Compton scattering ☐ D Nuclear fission

Scholarly Concept & Analysis: Mie scattering occurs when particle sizes equal or exceed light wavelengths, scattering all visible wavelengths equally to appear white.

Q18. What is the primary component of Photochemical Smog formed during sunny afternoons in polluted urban centers?

- ☐ A Pure water vapor ☐ B Ground-level Tropospheric Ozone (O_3) and Peroxyacetyl Nitrate (PAN) formed by UV reactions between NO_x and Volatile Organic Compounds (VOCs) ☐ C Frozen carbon dioxide ice ☐ D Inert helium gas

Scholarly Concept & Analysis: Photochemical smog is driven by solar UV light acting on vehicular NO_x and hydrocarbons, generating secondary toxic ozone and PAN.

Q19. Why is stratospheric ozone considered 'good ozone', while tropospheric ground-level ozone is considered 'bad ozone'?

- ☐ A They are completely different chemical molecules with different formulas ☐ B Stratospheric ozone shields terrestrial life from lethal UV radiation, while ground-level tropospheric ozone is a toxic, corrosive respiratory pollutant ☐ C Stratospheric ozone is breathable like pure oxygen ☐ D Tropospheric ozone is only found on Mount Everest

Scholarly Concept & Analysis: Ozone (O_3) in the stratosphere protects against ultraviolet rays; in the troposphere, it is a hazardous oxidant damaging human lungs and crops.

Q20. Which atmospheric layer reflects high-frequency skywave radio communications back to Earth due to free electrons and ions?

- ☐ A Troposphere ☐ B Ionosphere (part of the Thermosphere) ☐ C Stratosphere ☐ D Mesopause

Scholarly Concept & Analysis: The Ionosphere (D, E, and F layers) contains solar-ionized plasma that refracts and reflects HF radio waves for long-distance communication.

Q21. What is the 'Thermal Equator'?

- ☐ A A line drawn through active volcanoes ☐ B The fluctuating global belt around Earth that records the highest mean surface temperature at a given season, migrating north and south with solar declination ☐ C The geographic equator (0° latitude) exclusively ☐ D The boundary of the Earth's molten core

Scholarly Concept & Analysis: The thermal equator is the isotherm of maximum surface temperature, migrating seasonally into the Northern Hemisphere in July.

Q22. The 'Föhn' and 'Chinook' winds are examples of:

- ☐ A Cold sub-polar oceanic sea breezes ☐ B Warm, dry, katabatic/compressional leeward winds that descend mountain slopes, rapidly raising local temperatures and melting snow ☐ C Violent tropical maritime cyclones ☐ D Dense freezing smog clouds

Scholarly Concept & Analysis: Chinook and Föhn winds are warm, dry winds on leeward mountain slopes heated by adiabatic compression during descent.

Q23. What is the 'Walker Circulation' in equatorial meteorology?

- ☐ A A walking path across the Sahara desert ☐ B An east-west zonal atmospheric convective circulation over the equatorial Pacific Ocean driven by the temperature gradient between western and eastern Pacific waters
☐ C A polar jet stream circling the South Pole ☐ D A deep-sea tidal current in the Atlantic

Scholarly Concept & Analysis: The Walker circulation is the zonal atmospheric cell across the equatorial Pacific; its breakdown or displacement triggers El Niño.

Q24. How does aerosol pollution (such as sulfate particles and black carbon) exert a 'direct radiative forcing' on climate?

- ☐ A Sulfate aerosols scatter incoming solar radiation back to space (cooling effect), while dark black carbon absorbs solar radiation in the atmosphere (warming effect) ☐ B All aerosols double the Earth's magnetic field
☐ C Aerosols destroy the gravity of the Earth ☐ D Aerosols turn clouds into pure liquid nitrogen

Scholarly Concept & Analysis: Light-colored sulfate aerosols increase planetary albedo (cooling), whereas dark soot particles absorb radiation, warming atmospheric layers.

Q25. Which international agreement under the Montreal Protocol, adopted in 2016, mandates the global phase-down of hydrofluorocarbons (HFCs)?

- ☐ A The Kyoto Protocol ☐ B The Kigali Amendment ☐ C The Paris Agreement ☐ D The Copenhagen Accord

Scholarly Concept & Analysis: The Kigali Amendment (2016) to the Montreal Protocol mandates the phase-down of potent greenhouse-warming HFCs.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET G)

01 A B C D	02 A B C D	03 A B C D	04 A B C D	05 A B C D
06 A B C D	07 A B C D	08 A B C D	09 A B C D	10 A B C D
11 A B C D	12 A B C D	13 A B C D	14 A B C D	15 A B C D
16 A B C D	17 A B C D	18 A B C D	19 A B C D	20 A B C D
21 A B C D	22 A B C D	23 A B C D	24 A B C D	25 A B C D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & IN-DEPTH SCHOLARLY EXPLANATIONS (SET G)

Q. No.	Correct	Curricular Concept & Scholarly Analysis
1	C	Without the natural greenhouse effect, radiative equilibrium with solar insolation yields an effective emission temperature of 255 K (-18°C).
2	C	The dry adiabatic lapse rate ($\frac{g}{c_p}$) is approximately 9.8°C per 1,000 meters of elevation change.
3	B	When the environmental lapse rate exceeds the DALR, an ascending air parcel remains warmer and less dense than surrounding air, accelerating convection.
4	B	Geostrophic balance occurs above the friction layer when Coriolis force balances the pressure gradient force, flowing parallel to isobars.
5	B	The Southern Oscillation Index (SOI) measures normalized pressure anomalies between Tahiti and Darwin; sustained negative SOI indicates El Niño.
6	B	A positive IOD warms the western Indian Ocean relative to the east, reinforcing convective cells and boosting Indian monsoon rainfall.
7	B	The MJO is an eastward-propagating tropical wave of enhanced rainfall that modulates monsoon active-break cycles on 30–60 day timeframes.
8	A	Strong vertical wind shear tilts and disperses the warm latent heat core, disrupting the chimney effect and tearing the vortex apart.
9	B	Over a 100-year time horizon, methane traps approximately 28 to 36 times more infrared radiation per kilogram than CO_2 .
10	B	Sulfur hexafluoride (SF_6) is an extraordinarily potent greenhouse gas with a GWP of $\approx 23,500$ and an atmospheric lifetime of 3,200 years.
11	A	Ice loss replaces high-albedo ice with low-albedo ocean water, driving greater absorption of insolation and compounding warming.
12	A	The Clausius-Clapeyron equation dictates that the atmosphere's saturation vapor pressure expands exponentially by $\sim 7\%$ per °C of warming.
13	B	The transition period in October features clear sunny skies over soil still saturated with monsoon moisture, producing sultry 'October heat'.
14	B	Rossby waves arise from conservation of potential vorticity and the meridional gradient of the Coriolis force (β -effect).
15	B	The northward shift of the subtropical westerly jet beyond Tibet is a mandatory dynamic precursor to the onset of the Indian southwest monsoon.
16	A	Rayleigh scattering intensity is proportional to λ^{-4} ; short blue wavelengths scatter much more than longer red wavelengths, coloring the sky blue.
17	B	Mie scattering occurs when particle sizes equal or exceed light wavelengths, scattering all visible wavelengths equally to appear white.
18	B	Photochemical smog is driven by solar UV light acting on vehicular NO_x and hydrocarbons, generating secondary toxic ozone and PAN.
19	B	Ozone (O_3) in the stratosphere protects against ultraviolet rays; in the troposphere, it is a hazardous oxidant damaging human lungs and crops.

Q. No.	Correct	Curricular Concept & Scholarly Analysis
20	B	The Ionosphere (D, E, and F layers) contains solar-ionized plasma that refracts and reflects HF radio waves for long-distance communication.
21	B	The thermal equator is the isotherm of maximum surface temperature, migrating seasonally into the Northern Hemisphere in July.
22	B	Chinook and Föhn winds are warm, dry winds on leeward mountain slopes heated by adiabatic compression during descent.
23	B	The Walker circulation is the zonal atmospheric cell across the equatorial Pacific; its breakdown or displacement triggers El Niño.
24	A	Light-colored sulfate aerosols increase planetary albedo (cooling), whereas dark soot particles absorb radiation, warming atmospheric layers.
25	B	The Kigali Amendment (2016) to the Montreal Protocol mandates the phase-down of potent greenhouse-warming HFCs.